

Individual Reflection

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StudyOS Thesis-Finder • individual submission • University of Tübingen

1. Initial expectation

Given the course title, my initial expectation was a purely technical module where we build one large system against a goal defined by the professor and the whole difficulty sits in the implementation. What I did want out of it was to deepen my skills in working with coding agents, both on my own and in a team that uses these tools. The course turned out to work differently and that surprised me positively. In the first sessions we were forced to question what was even worth solving and how to approach the task at all, before writing a single line of code. Learning the tools happened as a byproduct of solving the assignment, not as a separate track. The framework that stuck with me is the double diamond. You have to question what is worth solving before you spend any thought on how to solve it, because in the age of coding agents the how has almost become secondary and the real bottleneck has moved upstream, to problem selection. At the beginning I assumed we would build a full assistant for students that helps with everything, especially learning. That is technically feasible, but it misses where the actual demand is.

2. Main learning

Building on that, my main learning was a much more concrete sense of how you assess which problems are worth solving. These are the same questions that matter in product discovery and in startups. Where is there demand, where could demand emerge and how do you evaluate that? Do other people really share my framing of the problem or am I just projecting my own view onto it? That is tied to actively asking for other people's opinions, because a single viewpoint on a problem is almost never complete. This clicked for me when we pivoted from building an app to building a skill. No user today wants to open a separate external tool just to find a thesis topic. A lightweight skill that plugs the task directly into an LLM the user already has open solves several of these problems at once. On top of that, the simple exercise of writing *we aim to solve X for Y by doing Z* and testing it against impact, feasibility and success already resolves a surprising number of open questions before they cost any implementation time.

3. Role and contribution in the group

We were a team of three with fairly similar contributions, but each of us leaned into a distinct strength. Max brought more expertise in building the app itself and in evaluating the skill's performance. Valentin focused on outward communication and on user testing through his own network. My role was mostly a meta-role in that setup. I repeatedly stepped back and asked whether what we were doing actually made sense and I regularly called for a pause to make the current state explicit and to clarify the next steps for the group.

The clearest example was the no-DB pivot. The group was split on whether to go database-backed or database-free and I pushed for no-DB. Instead of arguing it out, we built both in parallel for a short time. After the next group meeting it was clear that the two approaches perform similarly but that no-DB is significantly less effort in terms of long-term maintenance. My most concrete individual contribution was then following through on that direction and building the first no-DB prototype myself, which became the final artifact we shipped.

4. Future learning

Technically, I want to get better at working in larger AI-assisted teams: keeping communication clean, producing clean code and not letting errors creep in by holding the right balance between

tests and production code. I also want to sharpen my own vibe coding skill and find the right level of communication between me and the agent, so that code production runs efficiently while I still stay in control of the product instead of losing oversight of what is being built. Beyond that, the project grew my interest in building my own products. What appealed to me most was designing something with real demand behind it instead of something that is merely interesting to build. Together with seeing what coding agents are actually capable of, this pulled me towards working in or for a startup early in my career, because that is the environment where exactly these questions come up every day.